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Analyses of topical policy issues

The informal economy in Algeria: New insights using the MIMIC approach and the interaction with the formal economy[☆]

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ABSTRACT

This article aims first to investigate the main determinants of the informal economy (IE) in Algeria using the Multiple Indicator Multiple Causes (MIMIC) approach, second to estimate its size and development, third analyse the relationship in the short and long run between the informal and formal economy, and finally to explore the causality direction by employing the Granger causality test from 1980 to 2017. The results showed that the tax burden, the agricultural sector, the quality of institutions, and the GDP per capita are the main determinants of the IE in Algeria. The average IE size is 33.48% of official GDP, and it has been increasing in size over the past 15 years. Furthermore, it has been found that the IE positively affects the formal economy in the short run, while in the long run, this effect is reversed. Moreover, unidirectional Granger causality from GDP to IE is found. As for policy implications, in order to lower the size of IE in Algeria, the government needs to reconsider its tax policies, including the re-examination of policies implemented to stimulate the agricultural sector, which half of their effects are absorbed by the IE, and the generalization of the use of electronic transactions in economic activities.

1. Introduction

Informal economy (IE) is a significant issue for policymakers, particularly in emerging countries, where its existence is severe in comparison with developed countries (Schneider, 2005). IE falsifies formal indicators and increases the budget deficit by lowering government revenues, which may influence economic policies made by policymakers (Aigner et al., 1988). As a result, a growing number of studies have been devoted to analysing the determinants, estimating the size, and studying the interaction of the IE with other economic phenomena (Dell'Anno, 2003; Frey and Weck, 1983; Gutmann, 1977; Medina and Schneider, 2018). Despite the existing literature, scholars have been unable to agree on a single unified definition of IE; and, because every single definition is based on a different individual approach, various approaches to measuring the magnitude of the IE have been proposed.

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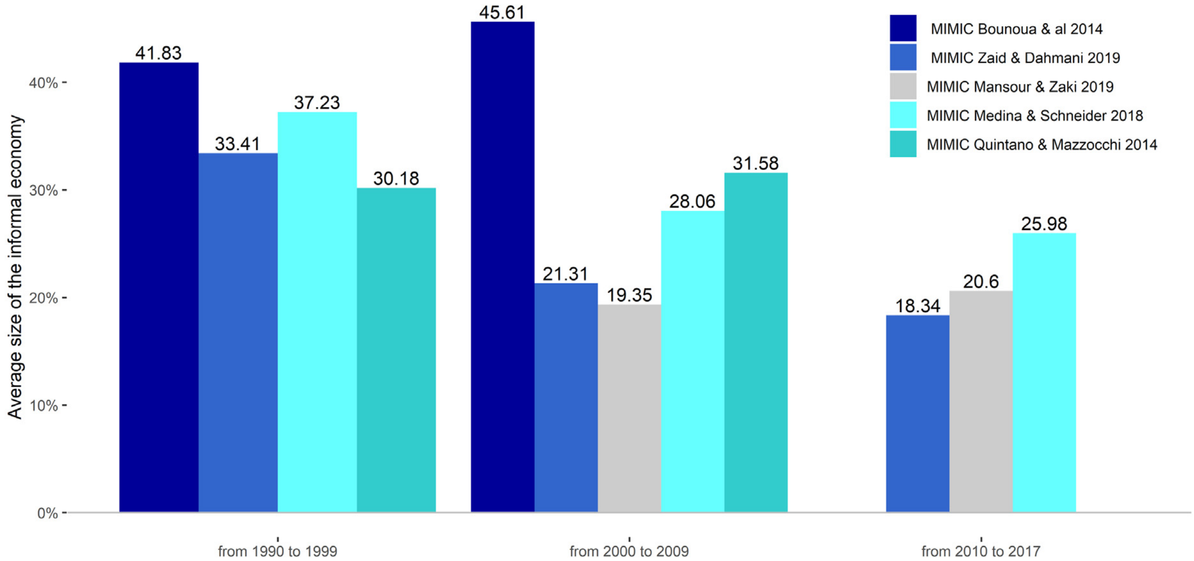

Fig. 1. Some existing estimates of the IE in Algeria using the MIMIC approach.

In recent decades, scholars have proposed numerous approaches in order to determine the causes of the IE in an attempt to predict the size and development of this complex economic phenomenon. Three main approaches can be distinguished in the existing literature. The first one is the direct approach, sometimes also referred to as a micro approach, which is based on sample surveys and audit measurements of undeclared taxable income (Wang et al., 2006). The second is an indirect approach. Unlike the previous one, this approach is based on macroeconomic data in general; the main idea of these methods is to measure traces left in formal statistics recorded. The third method is the model approach, which is based on structural equation modelling and the statistical theory of unobserved variables, implemented for the first time by Frey and Weck-Hanneman (1984). These approaches are applied worldwide; for example, Quintano and Mazzocchi (2014) used the model approach to estimate the IE for 17 Mediterranean countries and Hernandez (2009) adopted the Currency Demand Approach (CDA) in the case of Peru. The debate on which method is better or, in other words, more suitable to estimate the IE, is ongoing. However, according to Buehn and Schneider (2012), each approach has both advantages and disadvantages; hence, there is no ideal method to estimate the magnitude of IE.

Studies with estimates for the size of the Algerian IE as a percentage of the official GDP already exist (see Fig. 1). However, most of these studies address multiple countries (e.g., Abd El Aziz and Zaki, 2019; Alm and Embaye, 2013; Medina and Schneider, 2019) by employing a general model without considering the characteristics and specific features of each country, which may lead to omitting some significant causal variables or estimating different size and development behaviour. For instance, Medina and Schneider (2019) estimated the average size of the IE from 2000 to 2017 to be 32.2% of official GDP with a decreasing trend. In contrast, over the same period, Abd El Aziz and Zaki (2019) estimated an increasing trend with a slight drop from 2004 to 2008 and the average size of the IE to be 19.9% of official GDP. Furthermore, the two studies offered two different sets of causal variables that are assumed to influence the size of the Algerian IE. Based on these contradictions, there is a need to conduct more studies investigating the determinants and scope of this phenomenon explicitly in the Algerian case to help clarify the previous contradiction and improve the IE literature in Algeria, which is one of the main aims in this study.

From the existing literature, it is evident that there are a few studies devoted to analysing the IE's determinants and evolution exclusively in Algeria using indirect approaches, especially the model approach. The only exceptions are Bounoua et al. (2014) and Kori (2018). Unfortunately, these studies suffer from theoretical and econometrical issues (a more detailed discussion about these studies is provided in the second section). For example, Bounoua et al. (2014) chose to estimate the Multiple Indicator Multiple Causes (MIMIC) model using the Generalized Least Squares (GLS) estimator with a sample of 20 observations. However, this method requires a large sample because the estimated covariance matrix using GLS has a bias toward zero in a small sample (Bollen, 1989).

Furthermore, the best model selection was not based on the best of fit criteria or bed of fit criteria, but only on the R-squared and significance of individual coefficients. Moreover, the Kori's (2018) MIMIC model had only two determinants, rarely used in the literature, that is budget deficits and urbanization. More interestingly, the model results did not converge.

Consequently, and for the first time with special attention to the data characteristics (e.g., multinormality, stationarity), this article aims to analyse the determinants of the IE and estimate its size through a structural equation precisely the MIMIC approach from 1980 to 2017. In doing so, the authors adopt the definition of Hassan and Schneider (2016a, p. 2)

"The IE reflects mostly the legal economic and productive activities that, if recorded, should contribute to the national GDP". Thus, human trafficking, criminal activities, smuggling, and all related illegal pursuits are excluded. The MIMIC approach is selected for the following reasons: (a) although it has its critics, this approach is considered superior to the previous indirect approaches; (b) this approach treats the IE as an unobservable variable, which is consistent with the nature of this phenomenon; (c) it takes into account multiple causes and multiple indicators simultaneously, thus allowing the researcher to select a variety of determinants and indicators according to the characteristics of the economy being studied; (d) it is the most frequently used approach by academics and scholars to estimate the size of the IE.

In addition, and to the authors' best knowledge, no studies so far have empirically investigated the interaction between the informal and formal economy in Algeria, which is a literature gap we intend to contribute to its filling. Keeping in mind that the current studies, for example, Mughal and Schneider (2020), highlight the possibility of an asymmetric effect of the IE on the formal economy in the short and long run, we estimated a growth model using the Autoregressive Distributed Lag (ARDL) proposed by Pesaran et al. (2001). Besides, the Granger causality test is adopted in order to analyse the causality direction between the informal and formal economy.

To sum up, this article aims first to investigate the main determinants of the IE in Algeria using the MIMIC approach, second to estimate its size and development, third to analyse the relationship in the short and long run between the informal and formal economy, and finally to explore the causality direction by employing the Granger causality test from 1980 to 2017.

The rest of the paper is organized as follows. In Section 2, a theoretical background is provided, including the Algerian economy's structure, the existing estimation of the IE, and the interaction between the informal and formal economies. Section 3 is devoted to the MIMIC approach adopted to estimate the IE and the empirical results. Section 4 presents the empirical findings concerning the interaction between the formal and informal economies. The conclusion and policy implications are reported in Section 5.

2. Theoretical background

2.1. The economic structure of the Algerian economy

Before the discovery of oil in 1958, the Algerian economy was centred around the agricultural sector. Due to this discovery, Algeria has rapidly become the second-largest producer of oil and gas in Africa. After gaining political independence in 1962, Algeria joined the Organization of the Petroleum Exporting Countries (OPEC) in 1969. Two years later, Algeria declared the nationalization of hydrocarbons. Since then, hydrocarbon revenues have been regarded as the backbone of social, economic, and political life. In the 1980s, the Algerian government worked on reviving the industrial sector as an alternative to the hydrocarbons sector to achieve economic growth. During this time, the prevailing economic system in Algeria was the socialist system. In the late 1980s, the Algerian government abandoned it and adopted capitalism instead, as a consequence of internal and external pressures.

During the 1990s, the government introduced numerous economic reforms based on International Monetary Fund (IMF) and World Bank suggestions to restructure the economy and be compatible with the newly adopted market-based economic system. During this period, inflation and unemployment reached heights of 31.67% and 31.84%, respectively. However, the economic situation improved after the oil prices increased in 1998. Owing to the increase in the oil price in the late nineties, the Algerian government launched extensive programs of public investment to stimulate economic growth and improve the social welfare of the Algerian people.

Unfortunately, the high levels of corruption — according to Transparency International, Algeria ranked 106th out of 198 countries with a score of 65 in 2019 — had a significant impact on the performance of investment programs at the beginning of the 21st century. In this period, the most famous scandals involved the bribery of major officials in the Sonatrach company (a nationally owned oil company founded in 1963) that reached USD 250 million. In addition to this, USD 11 billion was wasted in the East–West Highway infrastructure project.

As aforementioned, dependence on the hydrocarbons sector is still ongoing. For instance, before the global pandemic, the contribution of the hydrocarbons sector reached 22.78% of GDP and more than 85% of export earnings. As a consequence, the Algerian economy is affected by fluctuations in international oil prices.

Overall, the Algerian economy has gone through many political and economic structural reforms, which lead to a troubled economy vulnerable to oil price shocks. In addition, high levels of corruption have made the Algerian government incapable of achieving important social and economic development. As a result of the disappointing performance of the formal economy, many people turned to the IE in Algeria to seek alternative sources of revenue (CNES, 2004). The following sub-section provides an overview of existing estimations of the IE in Algeria.

2.2. Existing estimates of the IE in Algeria using macroeconomic approaches

Studies that have assessed the size of the IE in Algeria via indirect approaches have been relatively small (Alm and Embaye, 2013; Boudlal, 2012; Bouriche and Bennihi, 2020), especially those using the model approach (Abd El Aziz and Zaki, 2019; Bounoua et al., 2014; Buehn and Schneider, 2012). Table 1 below summarizes studies that have assessed the volume of the IE in Algeria, and it is noted that most of these studies are not devoted exclusively to Algeria.

Table 1
Existing estimates of the IE in Algeria.

Authors	Study period	Methodology	Estimation (average)
Country-based estimates			
Boudlal (2012)	1970–2010	CDA	24.5% of GDP
Bounoua et al. (2014)	1990–2009	MIMIC	44.72% of GDP
Bouriche and Bennihi (2020)	1980–2019	CDA	21.% of GDP
Panel-based estimates			
Elgin and Oztunah (2012)	1960–2008	DGE	33.09% of GDP
Alm and Embaye (2013)	1994–2006	CDA	48.09% of GDP
Quintano and Mazzocchi (2014)	1995–2010	SEM	39.4% of GDP
Medina and Schneider (2018)	1991–2015	MIMIC	30.86% of GDP
Abd El Aziz and Zaki (2019)	2000–2017	MIMIC	19.98% of GDP

Notes: CDA: Currency Demand Approach. MIMIC: Multiple Indicators Multiple Causes approach. DGE: Dynamic General Equilibrium.

Table 2
Existing estimates of the IE in Algeria.

	MIMIC approach				CDA approach	
Estimations	B-S	M-S	Q-M	A-Z	A-E	B-B
B-S	1					
M-S	−0.4112 [0.091]	1				
Q-M	0.0969 [0.702]	**−0.5162** **[0.028]**	1			
A-Z	−0.3174 [0.185]	**0.9526** **[0.000]**	**−0.5995** **[0.01]**	1		
A-E	0.3073 [0.308]	0.4791 [0.1]	−0.3610 [0.225]	0.5347 [0.217]	1	
B-B	**−0.5986** **[0.001]**	0.3132 [0.127]	**−0.507** **[0.03]**	−0.3725 [0.128]	0.1054 [0.732]	1

Notes: *p*-value in brackets. CDA: Currency Demand Approach; B-S: Bounoua et al. (2014); M-S: Medina and Schneider (2018); Q-M: Quintano and Mazzocchi (2014); A-Z: Abd El Aziz and Zaki (2019); A-E: Alm and Embaye (2013); B-B: Bouriche and Bennihi (2020).

In a panel study by Quintano and Mazzocchi (2014), it was found that the average size of the IE was 39.4% of official GDP from 1995 to 2010. However, the authors used a unified model to estimate the IE in a heterogeneous sample consisting of industrialized and emerging countries. They did not account for differences in historical background, geographical distance, economic characteristics, and cultural roots, which can bias estimations. Unlike the previous study, Medina and Schneider (2018) took these differences into serious account. The empirical investigation showed that the IE was decreasing from 1991 to 2015, with an average of 30.86% of official GDP and a standard deviation of 5.47%. A more recent study made by Abd El Aziz and Zaki (2019) estimated the IE to be 19.9% from 2000 to 2017.

When compared (see Table 2), these estimations do not converge to the same result. For example, the estimations of Medina and Schneider (2018) are negatively correlated with a 5% significance level with Quintano and Mazzocchi (2014) and positively associated with Abd El Aziz and Zaki (2019) at a 1% significance level. At the same time, the estimations of Quintano and Mazzocchi (2014) are negatively correlated with the estimations of Abd El Aziz and Zaki (2019) with a 5% significance level. Furthermore, the latter estimations are not correlated with the estimations of Alm and Embaye (2013).

The studies devoted exclusively to analysing the Algerian case were Bounoua et al. (2014) and Kori (2018). The first study analysed the evolution of the Algerian economy and concluded that the main causes of the IE were public expenditure, inflation rates, unemployment rates and the guaranteed national minimum wage from 1991 to 2009. The second study estimated the size of the IE from 1970 to 2016 and found that budget deficits and urbanization are the main determinants. However, for both studies, some critical remarks can be made concerning their theoretical and methodological aspects. Bounoua et al. (2014) chose to estimate the MIMIC model using the GLS estimator with a sample of 20 observations. According to Bollen (1989), this method requires a large sample because the estimated covariance matrix $\hat{\theta}$ using GLS has a bias toward zero with a small sample. The authors also selected the optimal model using coefficients' significance and R-squared rather than the goodness of fit indexes. Kori's (2018) model includes only two determinants rarely used in the literature budget deficits and urbanization. Another interesting point is that the author chose to constrain GDP to one. According to Schneider (2005), this relationship exists only in developed countries which is not the case for Algeria. Furthermore, the model did not converge, meaning that the estimated coefficients are not reliable.

Due to the inconsistencies in the panel MIMIC estimations and shortcomings of studies devoted exclusively to Algeria, more investigation is needed to reveal the properties of the Algerian IE, which is conducted in the following parts.

2.3. Interaction between informal and formal economy

The complex and continuous dynamic interaction between the IE and the formal economy makes it very hard to separate the two economies from each other. In general, the IE influences the formal economy through three channels, namely taxation, biased effects of economic policies, and general allocations. Findings differ between countries. A relationship between the two sectors is not conclusive, and it is unclear whether negative effects predominate over positive effects or vice versa.

On the one hand, Loayza (1996) constructed a simple endogenous growth model with two main assumptions; the first is that production technology relies principally on congested public services; the second is that governments are unable to enforce full compliance and tend to impose excessive taxes and regulations. As a result, the author found that the IE has a negative impact on the growth of the formal economy under weak enforcement of compliance and the statutory tax burden is bigger than the optimal tax burden. Furthermore, many empirical studies support this finding and have concluded that the IE harms formal economic growth. These studies include those of Schneider et al. (2010) in the case of OECD countries, Schneider and Hametner (2014) in the case of Colombia, and Soares and Afonso (2019) in the case of Portugal.

On the other hand, Adam and Ginsburgh (1985) found a positive impact when analysing the impact of the IE on economic growth in the case of Belgium under certain assumptions such as low entry costs into the IE due to low likelihood of enforcement. More recent studies by Mapp and Moore (2015) in the Caribbean region, and Zamman and Goschin (2015) in the case of Romania, also produced the same results empirically.

Furthermore, in the view of Hassan and Schneider (2016b), the level of development determines the type of this effect. That is, in developed countries, the positive effect is dominant, while in developing countries, it is the negative effect that dominates.

3. Empirical estimates of the IE in Algeria

3.1. Data, method of estimation, and choice of variables

The data are taken from Algeria's official national statistics, except for the bureaucracy quality index, which is extracted from the International Country Risk Guide (ICRG) database. The data retrieved are an annual macroeconomic time series that covers the period from 1980 to 2017. This period was chosen due to data availability. Only GDP and currency to M1 ratio were log-transformed to overcome the problem of outliers found in these series.

The model approach is used to define the key determinants of the IE in Algeria, which relies on the statistical theory of latent variables and structural equation modelling, particularly the MIMIC model. This approach is selected in comparison to other proposed methods in the literature (direct and indirect approaches) to determine the causes and indicators of the Algerian IE for several reasons: (a) although it has its critics, this approach is considered superior to the previous indirect approaches; (b) this approach treats the IE as an unobservable variable, which is consistent with the nature of this phenomenon; (c) it takes into account multiple causes and multiple indicators simultaneously, thus allowing the researcher to select a variety of determinants and indicators according to the characteristics of the economy being studied; (d) it is the most frequently used approach by academics and scholars to estimate the size of the IE.

3.1.1. Method of estimation

The MIMIC model belongs to the Linear Interdependent Structural Relationships (LISREL) models. Jöreskog and Goldberger's (1975) MIMIC model mainly consists of two equations: the first is the measurement equation, which links the unobserved variables to a set of manifested indicators.

$$\eta = \gamma' x + \zeta \tag{1}$$

Where η is the latent variable, x is a $(1 \times q)$ vector of observable causal variables, γ is $(1 \times q)$ vectors of structural parameters, and ζ is the error term.

The second equation specifies the causal association between the unobserved variables, and it is referred to as the structural equation. In the current paper, there is one unobserved variable, the IE size. It is supposed to be indirectly observable by a set of indicators of the latter, thus catching the structural dependence of the IE on variables.

$$y = \lambda \eta + \varepsilon \tag{2}$$

Where y is a $(1 \times p)$ vector of observed indicator variables, λ is a $(1 \times p)$ vector of regression parameters, and ε is a $(1 \times p)$ vector of measurement error term assumed to be white noise. It must be noted that the assumption of independence between structural disturbance ζ, and measurement error ε is central to the reliability of estimates.

The advantages of the MIMIC come from considering multiple indicators instantaneously. According to Tedds (1998), in the case of the IE, it is obvious that the effects are likely to show up simultaneously in numerous markets, including the production market, labour market, and monetary market. In addition, the causes are more numerous and more complex than just the tax rate. Moreover, this approach allows the researcher to select a variety of determinants and indicators according to the study objective, the characteristics of the economy being studied and the availability of the data (Giles, 1999).

However, this approach has some downsides when estimating the IE. The researcher is more likely to determine whether a certain model is valid than to find an appropriate model, as the MIMIC approach is a confirmatory rather than exploratory technique. Another criticism made by Breusch (2005) and Helberger and Knepel (1988) concerns the stability of the estimated coefficients. They argue that changes in sample size or units of measurement lead to different estimations. Besides this, the researcher has to choose a suitable calibration method because the model only produces an ordinal series. To overcome this last problem, scholars have proposed several benchmarking or calibration methods to convert the IE index to a series of percentages of official GDP.

Despite these downsides, the approach is still superior compared to other approaches. The adopted approach has been applied by economists (e.g., Macias and Cazzavillan, 2010; Mughal and Schneider, 2020; Zamman and Goschin, 2015) to estimate the size of the IE in recent decades and is still used today.

3.1.2. Choice of variables

Choosing which variables to include in this study is a great challenge, especially with many empirical studies finding some controversial results. Therefore, and to achieve objectiveness, over a hundred studies that investigated the determinants and scope of IE were gathered and analysed to retrieve information on the used causal variables and indicators to reflect the IE size. After this exercise, it was found that scholars have used different combinations of IE causes and indicators. The main determinants in previous studies have been tax burden, unemployment rate, government intervention in the economy, self-employment, and the quality of institutions. Besides these causes, economists have proposed others based on the characteristics of the country being studied. In the current study, tax burden, the share of the agricultural sector, the status of the formal economy reflected by unemployment and GDP per capita, and quality of institutions are considered as causes of the IE, in addition to GDP, money supply and employment as indicators. The theoretical justifications for choosing these variables are discussed below.

3.1.2.1. Causal variables.

(1.1) Tax burden: In the literature, a rise in the tax burden stimulates a strong incentive to engage in IE activities (Buehn and Schneider, 2012), and therefore a positive sign is expected. To investigate this point, this variable is reflected as the aggregation of direct and indirect taxes normalized by GDP as a proxy for the tax burden. The following hypothesis is developed:

Hypothesis 1. The higher the tax burden, the higher the IE, ceteris paribus.

(1.2) The size of the agricultural sector*:* According to Vuletin (2008), the size of the agricultural sector and IE are positively correlated due to feeble control and governance, especially in rural areas. This relationship is supported by many studies around the world, particularly in developing countries (Angour and Nmili, 2019; Hassan and Schneider, 2016a).
Therefore, we include this variable in our study as the value added by the agricultural sector normalized by the GDP. To this end, the following hypothesis is developed:

Hypothesis 2. The larger the agricultural sector, the higher the IE, ceteris paribus.

(1.3) Quality of institutions: In literature, several studies link institutional quality to the IE, especially in developing countries (Loayza, 1996; Schneider et al., 2010). According to Medina and Schneider (2018), the impact of this variable on the IE depends on the level of corruption in the country; in other words, a less corrupt bureaucracy has a smaller IE (Bovi and Dell'Anno, 2010; Torgler and Schneider, 2007), while high levels of corruption tend to reverse this relation.
In the Algerian case, there is a literature gap on this matter; to our best knowledge, this variable has not been included in studies concerned with the scope and dynamics of the IE of Algeria. Therefore, the quality of institutions is included in the study model. The two scenarios above are investigated in the case of Algeria, where corruption levels are high. The latter is measured by the bureaucracy quality index from the ICRG database. The following hypothesis is developed:

Hypothesis 3. The better the quality of institutions, the lower the IE, ceteris paribus.

(1.4) Per capita income: According to Quintano and Mazzocchi (2014), individuals in developing countries are more likely to work in the IE to survive. Thus, a negative association is found. Buehn and Schneider (2012) give another explanation for this relationship; that is, when an economy is booming, individuals are more motivated to work in the formal economy rather than the IE. In another study by Ángel et al. (2005), a positive relationship between individual income and the size of the IE is found. They argue that higher income stimulates individuals to buy goods and services from both informal and formal economies.

Due to the high interference between the two economies in Algeria, the GDP per capita growth rate is added to better capture the influence of the IE on the formal economy in Algeria. For this variable, the following hypothesis has been developed:

Hypothesis 4. The higher the GDP per capita, the lower the IE, ceteris paribus.

(1.5) Unemployment rate: Despite the existing literature, the association between the IE and unemployment is still ambiguous. Some studies have found a positive association. According to Macias and Cazzavillan (2010), this is because unemployment stimulates individuals to search for an income in the IE. Nevertheless, a handful of studies have found a negative association, for example, Corina and Andrie (2011) in Romania and Macias and Cazzavillan (2010) in the case of Mexico. This association is explained by the generalized adverse economic situation, which should or could affect both informal and formal economies equally. Thus, a negative association is detected.
In the Algerian context, this relationship between unemployment and IE is a point for further empirical analysis because previous studies were inconclusive. For this variable, the following hypothesis has been suggested:

Hypothesis 5. The higher the unemployment, the larger the IE, ceteris paribus.

3.1.2.2. Indicator variables.

(2.1) Development of production market (log current GDP): When using Structural Equation Modelling (SEM), it is necessary to fix a scale variable to estimate the remaining coefficients as a function of this scale variable. In most studies in the literature, this variable is the formal economy measured by GDP. However, the relationship between the formal economy and IE is still ambiguous; some studies found a positive correlation, while others found a negative association.
In our case, the log current GDP is used. Based on Schneider et al. (2010), in developing countries, this association is negative. Therefore, this variable is fixed to −1, and we follow Dell'Anno (2003) strategy to determine the correct sign of the association between the formal economy and the IE in Algeria. For this variable, the following hypothesis has been developed:

Hypothesis 6. The lower the IE, the higher the log GDP, ceteris paribus.

(2.2) Development of monetary market: Given that individuals engaged in the IE do not want to leave any traces leading back to their activities in this economy, thus all their transactions in this economy are made in cash. Accordingly, a positive sign is expected.
Consequently, the current study uses currency to M1 as an indicator of the size of the Algerian IE to take the previous point into account. For this variable, the following hypothesis is suggested:

Hypothesis 7. The lower the IE, the less currency is held by the public, ceteris paribus.

(2.3) Development of the labour market: Many studies have considered the workforce participation rate as an indicator (Corina and Andrie, 2011; Dell'Anno et al., 2018). In our case, this variable is calculated by normalizing the total labour force to the working-age population (15–64 years old). In economic literature, this association is found to be positive. Based on this, the current study investigates this relationship in Algeria by testing the following hypothesis:

Hypothesis 8. the lower the IE, the higher employment, ceteris paribus.

Therefore, as shown in Fig. 2, the theoretical model of the current study consists of five manifested causes, three manifested indicators, and one latent variable, which is the IE. Thus, in general, the theoretical study model is (MIMIC 5-1-3). In addition to the above causes, several other determinants were included, such as government spending, inflation, urbanization rate, and a dummy variable to capture the impact of the transition from socialism to capitalism. However, they were not significant.

Several indexes were used to assess the quality adjustment in the model estimated. The goodness of fit indicators used were chi-square, Comparative Fit Index (CFI), and the Tucker–Lewis index (TLI). Concerning the last two indexes, values closer to one indicate a good fit. Standardized Root Mean Square Residual (SRMR) and Root Mean Square Error of Approximation (RMSEA) were used to assess the badness of fit. According to Hu and Bentler (1999), values lower than 0.08 indicate good adjustment.

3.2. Estimation results

Before making any further empirical analysis, the statistical properties of the data needed to be checked. Table 3 provides concise statistics of the time series used in the study. It is noted that the mean and median are very close across variables, except for the unemployment rate. The latter variable also has the longest range in the dataset. The values of skewness and kurtosis are found to be in the expected range for all variables. The Shapiro–Wilk univariate normality test reveals that only the added value of the agricultural sector and per capita income are normally distributed (p-value > 0.05).

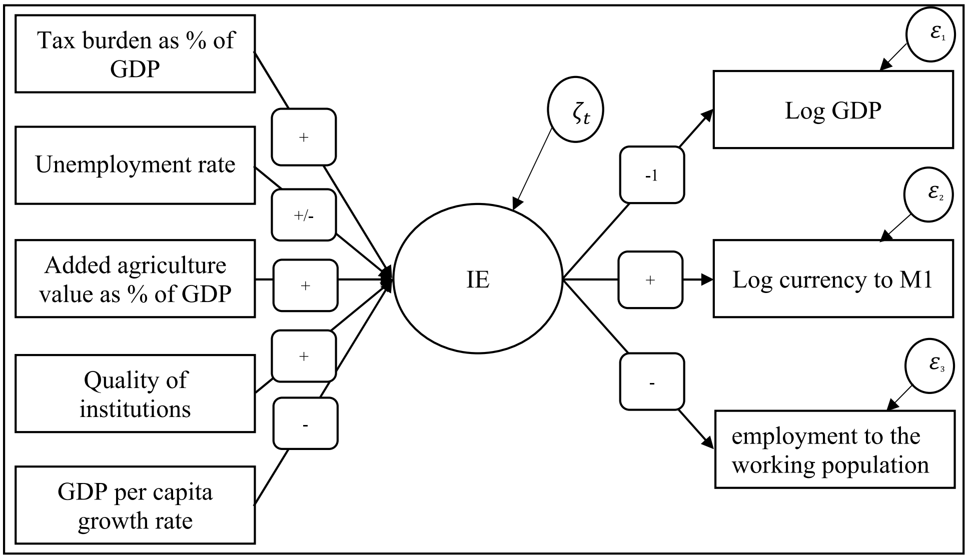

Fig. 2. Theoretical model MIMIC (5-1-3).

Table 3
Descriptive statistics of the data from 1980 to 2017; yearly observations.

Variables	Mean	Median	Max	Min	Skew	Kurtosis	S-W	Obs.
Determinants								
Tax burden as % of GDP	3.87	3.01	6.49	1.99	0.44	1.68	0.874	38
Unemployment rate	18.34	15.71	31.84	9.82	0.32	1.60	0.890	38
Added agriculture value as % of GDP	9.52	9.59	12.27	6.57	0.08	1.97	0.965	38
Quality of institutions	1.70	2.00	3.00	1.00	0.08	2.37	0.769	38
GDP per capita growth rate	0.69	1.25	5.84	−4.25	−0.33	2.63	0.959	38
Indicators								
Log current GDP	28.49	28.74	30.55	25.74	−0.28	1.64	0.904	38
Log currency to M1	3.76	3.81	3.92	3.41	−0.73	2.55	0.893	38
Employment to the working population	29.97	29.60	39	22.51	0.17	1.60	0.916	38

Notes: Data are in level. S-W is the Shapiro–Wilk test for normality, the critical value is W = 0.938 at 5% significance. Obs.: observations

3.2.1. Stationarity tests

Macroeconomic time series usually exhibit non-stationarity problems, which may lead to spurious regression. To avoid this, the study proceeds to investigate non-stationarity by using a battery of unit root tests, including traditional time series unit root tests such as the Augmented Dickey and Fuller (1979) (ADF) and Phillips and Perron (1988) (PP). In addition, the Zivot and Andrews (2002) (ZA), which allows for breaks at a point in the intercept and/or the trend to overcome the shortcomings of the traditional unit root tests used.

Table 4 shows the results of stationarity properties based on ADF, PP and ZA unit root tests. ADF and PP show that only the per capita income is significantly stationary at a level of 5%, while all other variables become significantly stationary after the first difference. The results of the break test of ZA corroborate the integration status of the variables. The reported time breaks fall between 1990 and 2003 for all series; this coincides with a transition from a socialist economy to capitalism and the policies imposed by the International Monetary Fund (IMF) in the early 1990s. Depending on the previous results, all variables are transformed by taking the first difference except for per capita income.

3.2.2. Results of the MIMIC model

After inspecting the stationarity and determining the integration order of the time series, the next step is to estimate the MIMIC model. Several models were taken into consideration and tested to select the best one, starting with the most general theoretical model (MIMIC 5-1-3) and omitting non-significant variables. Four of these models are presented in Table 5. The reported coefficients are in the standard form, and for every specification, the multi-normality assumption is tested using Mardia (1974) and Henze and Wagner (1997) multi-normality tests. The results indicate that the data does not satisfy the multi-normality assumption. Hence, robust maximum likelihood is applied.

Based on the selection criteria, model 4 (MIMIC 4-1-2) is considered to be the optimal model. As shown in Table 5, RMSEA is estimated to be 0.07 with a *p*-value greater than 5% and the SRMR is found to be 0.051, which means that the

Table 4
Unit root tests over the period from 1980 to 2017.

Variables	At levels					
	ADF		PP		ZA	
	I	I&T	I	I&T	I&T	Break
Tax burden as % of GDP	1.256	1.118	−3.985	−4.047	−2.984	1998
Unemployment rate	0.6347	1.987	−2.802	−3.579	−3.001	1991
Added agriculture value as % of GDP	1.563	1.452	−10.18	−10.542	−3.464	2004
Quality of institutions	7.979***	11.59***	−10.584	−13.553	−3.860	1990
GDP per capita growth rate	4.269	4.219	−23.325***	−23.567***	−5.982***	1994
Log current GDP	5.348	1.041	−0.715**	−0.879**	−2.478	1990
Log currency to M1	2.422	2.374	−8.639	−8.578	−4.161	2003
Employment to the working population	1.475	2.816	−4.287	−15.164	−4.560	1999
	First difference					
	ADF		PP		ZA	
	I	I&T	I	I&T	I&T	Break
Tax burden as % of GDP	8.043***	8.126**	−27.363***	−27.313***	−6.595***	1990
Unemployment rate	5.726**	6.449*	−31.024***	−31.834***	−6.212***	1999
Added agriculture value as % of GDP	14.456***	14.123***	−44.227***	−44.409***	−8.754***	1988
Quality of institutions	11.945***	11.691***	−17.450**	−17.458*	−4.82	1988
GDP per capita growth rate	18.639***	17.930***	−42.091***	−42.765***	−8.949***	1986
Log current GDP	6.008**	7.559**	−21.284***	−22.424***	−5.559**	1987
Log currency to M1	10.340***	9.891***	−30.094***	−31.442***	−5.853***	1985
Employment to the working population	11.577***	11.212***	−42.921***	−42.925***	−7.771***	2002

Notes: ***, **, * is the significance level at 1%, 5% and 10%, respectively. I represent the model with intercept only, and I&T represents the model with an intercept and trend. ADF is Augmented Dickey–Fuller Unit Root Test, the number of lags was determined using AIC criteria for maximum 12 lags to remove serial correlation in the residuals. PP is the Phillips–Perron unit root test, the test was conducted using Newey–West Bandwidth. ZA is Zivot–Andrews unit root test, lag selection was based on t-statistic.

Table 5
Model estimation results over the period of 1980 to 2017.

	Model 1 (5-1-3)	Model 2 (4-1-3)	Model 3 (5-1-2)	Model 4 (4-1-2)
Determinants				
Tax burden as % of GDP	0.523***	0.516***	0.505***	0.499***
	(4.562)	(4.42)	(4.475)	(4.354)
Unemployment rate	0.0104		0.085	
	(0.939)		(0.749)	
Added agriculture value as % of GDP	0.534***	0.556***	0.477***	0.494***
	(5.507)	(6.357)	(4.79)	(5.544)
Quality of institutions	0.365***	0.345***	0.361***	0.344***
	(4.693)	(4.568)	(4.573)	(4.562)
GDP per capita growth rate	−0.305**	−0.286***	−0.284**	−0.269**
	(−3.034)	(−2.891)	(−2.749)	(−2.648)
Indicators				
Log current GDP	−1	−1	−1	−1
Log currency to M1	0.39**	0.377**	0.347**	0.34**
	(2.460)	(2.459)	(2.28)	(2.296)
Employment to the working population	0.222*	0.22*		
	(1.807)	(1.768)		
Model fit indicators				
Chi-square	18.71	16.74	6.28	4.73
	[0.07]	[0.05]	[0.28]	[0.32]
RMSEA	0.14	0.152	0.085	0.07
	[0.10]	[0.08]	[0.33]	[0.27]
SRMR	0.08	0.08	0.059	0.051
CFI	0.845	0.848	0.968	0.98
TLI	0.747	0.747	0.929	0.95
Degrees of freedom	26	19	20	14
R-square	0.865	0.864	0.761	0.76

Note: ***, **, * is the significance level at 1%, 5% and 10%, respectively. Z-value in parentheses. P-value in brackets. Degrees of freedom $= 0.5(p+q)(p+q+1)-j$, where p is the number of causal variables, q is the number of indicator variables, j is the number of free parameters.

two indicators are within the accepted range. The goodness of fit indicators CFI and TLI are estimated to be 0.99 and 0.97, respectively, which means that the model is well fitted.

As displayed in Table 5, all the included variables have the right expected sign across the four models, and most of them are highly significant, except for unemployment and the ratio of employment to the working population, which is significant at the 10% level.

Starting with the determinants, the tax burden is found to impact the size of the IE positively. This impact is significant at the 1% level across all models. In the selected model, a one per cent increase in the tax burden increases the size of the IE by 0.50%, ceteris paribus. The results were found to support Boudlal's (2012) findings concerning the impact of the tax burden on the size of the IE in Algeria. According to Buehn and Schneider (2012), the difference between total labour cost in the formal economy and earnings after subtracting taxes is a key factor in motivating individuals to work in the IE. Thus, higher taxes on income motivate individuals to engage in IE's activities.

The second most important cause is the agricultural sector. This factor is found to influence the IE positively. This impact is significant at the 1% level across the four specifications. A one per cent increase in the agricultural sector increases the IE by 0.49%, ceteris paribus. As in the cases of Egypt, Morocco and Yemen (Angour and Nmili, 2019; Medina and Schneider, 2018), the Algerian IE is also positively affected by the agricultural sector. In recent years, the Algerian government has been trying to develop economic activities outside the hydrocarbons sector, and agriculture is one of the important sectors that the government is relying on to reduce dependency on oil exports. Unfortunately, the results indicate that stimulating the agricultural sector led to growth in the IE; this is because the majority of workers in this specific sector are informal workers, and there is no formal mechanism for managing them in this sector, in addition to the difficulties imposed by bureaucracy in operating it.

Another important cause is the quality of institutions. Ceteris paribus, a one per cent rise in the latter increases the IE by 0.34%. As stated before, due to the high levels of corruption in Algeria, this variable has a positive and significant impact at the 1% level. This result was found by Dobre and Adriana (2009) in the case of Japan, which also has a significant level of corruption.

For the causes reflecting the state of the formal economy, GDP per capita is found to have the expected sign in all models, and it is significant at the 5% level. The same result is found in Medina and Schneider (2019). As suggested, an increase in GDP per capita decreases the IE. In Algeria, this impact is estimated to be 0.27%. This result means that individuals' engaging in informal activities depends on the situation of the IE. In times of recession, individuals attempt to regain their lost revenue from official activities by engaging in the IE. However, when the economy is recovering, these individuals will be less engaged in informal activities due to opportunities to earn income in the formal economy.

Unemployment is found to have a positive but non-significant coefficient across all models. This result means that the IE in Algeria is not affected by unemployment. This weak statistical significance is found when the labour force in the IE is heterogeneous. A closer look at this labour force reveals that some individuals are officially unemployed; others are housewives who are not included in the official workforce, retired people, immigrants, and individuals who work in the formal economy and IE at the same time (Torgler and Schneider, 2007; Wang et al., 2006), which is consistent with the Algerian case. Looking at the history of unemployment in Algeria, it is noted that it was not stable, especially in the 1990s. This is due to the transition from the socialist system to the capitalist system and the economic reforms adopted by the Algerian government, which led to the closure of many factories and the laying off of nearly 2000 workers.

Furthermore, this period witnessed social and political instability, which complicated the situation even more. Since 2000, the Algerian government has reduced the unemployment rate by recruiting individuals in the public sector; this policy has reduced unemployment rates in the short term. However, it is not very effective in the long run because the public sector cannot keep up with the growth of the workforce. Furthermore, the policies adopted to balance the demand for work and job offers are ineffective. In addition to degradation in purchasing power, this has led people to participate in the IE.

For the indicators, the two indicators have the right sign. However, only log currency to M1 is statistically significant at the 5% level. A one per cent increase in the size of the IE increases cash demand by roughly 34%, which confirms that individuals use cash in conducting their transactions in the IE.

The above results are summarized in Table 6, where all hypotheses are confirmed. However, the second hypothesis, which states that the increase in unemployment rates will lead to a rise in the IE, is inconclusive because it lacks statistical significance; this can be interpreted as follows: "the rise in unemployment rates does not necessarily raise the IE in Algeria". The same result was found by Hassan and Schneider (2016a) in the case of Egypt. Moreover, Medina and Schneider (2019) concluded that the unemployment rate is a less influential causal variable for developing countries. Although the established sign is in line with the previous studies, further investigation is needed.

3.2.3. The size of the Algerian IE

Based on the selected MIMIC model, the structural equation of the selected model is written as:

$$\frac{\Delta\tilde{\eta}}{GDP_{2002}} = 0.499\Delta Tax\ burden + 0.494\Delta Agriculture + 0.344\Delta Quality\ of\ institutions - 0.269\Delta Per\ cap\ incom \tag{3}$$

The index of IE is calculated by structural equation (3), and since the determinant variables are in the first difference, the unobserved variable is estimated at the same level.

Following the procedure of Dell'Anno et al. (2007), the size of the IE is calculated using Eq. (4). However, this procedure needs an external value to get real values of the IE as a percentage of formal GDP. For this, a year that satisfies two

Table 6
Results of hypothesis testing.

Hypothesis	Expected sign	Result
Hypothesis 1	Positive	Confirmed
Hypothesis 2	Positive/negative	Positive insignificant
Hypothesis 3	Positive	Confirmed
Hypothesis 4	Positive	Confirmed
Hypothesis 5	Negative	Confirmed
Hypothesis 6	Negative	Confirmed
Hypothesis 7	Positive	Confirmed
Hypothesis 8	Negative	Confirmed

conditions was selected in order to assure truthfulness. The first condition is the availability of several estimates, and the second is that they are close to each other. The base year chosen is 2002, where the size of the IE is estimated to be 31.9% of official GDP (Medina and Schneider, 2018).

$$\frac{\tilde{\eta}_t}{GDP_{2002}}\left[\frac{\eta^*}{GDP_{2002}}\times\frac{GDP_{2002}}{\tilde{\eta}_{2002}}\right]\frac{GDP_{2002}}{GDP_t}=\frac{\hat{\eta}_t}{GDP_{2002}} \tag{4}$$

Where:
$\frac{\tilde{\eta}_t}{GDP_{2002}}$ is the index calculated by Eq. (3).
$\frac{\eta^*}{GDP_{2002}}=31.9\%$ is the external estimation obtained from Medina and Schneider (2018).
$\frac{GDP_{2002}}{\tilde{\eta}_{2002}}$ is the value estimated by Eq. (4) in the year 2002.
$\frac{GDP_{2002}}{GDP_t}$ is able to convert the index of IE as change respect to the base year in respect to current GDP.
$\frac{\hat{\eta}_t}{GDP_{2002}}$ is the estimated IE as a percentage of official GDP.

Fig. 3 shows the estimation results compared to some previous estimations of the IE in Algeria using the MIMIC model. As shown, the current estimation has the longest period, which is from 1980 to 2017. The plot shows that the size of IE started to rise from 2005, except for the estimations of Quintano and Mazzocchi (2014).

Fig. 4 represents the estimated size of the IE in Algeria from 1980 to 2017. It is noted that the IE during this period varies between 31% and 37%, with an average of 33.48%, with the highest percentage in 2017 and the lowest in 2005. Furthermore, for the properties of the Algerian IE to be analysed, the strucchange package made by Zeileis et al. (2002) is used, which implements the algorithm described in Bai and Perron (2003). The results suggest five main breakpoints, two negative shocks and three positive shocks. The two negative shocks reduced the IE from 1994 to 2007. This decrease coincided with two major structural reforms in the tax system, the first in 1992 and the second in 1994. As for the three positive shocks, the first was in 1985 in conjunction with the oil price shock, while the second and third were consecutive, starting from 2007 in conjunction with the 2008 economic crisis until 2017. The MIMIC estimation is in line with our previous estimation (Bouriche and Bennihi, 2020) using the CDA, with a positive correlation of 60% significant at the 1% level.

3.2.4. Robustness check

Several additional analyses were performed to check the robustness of the results. These tests included: (1) a stability analysis, in which the period was varied by dropping observations at the beginning and end of the study period; (2) estimating the selected model without GDP per capita as a causal variable over the original study period; (3) re-estimating the four models using standard data with a mean of zero and standard deviation of one and using the variables in growth form.

First, the stability of the results was tested by dropping observations at the beginning of the period; the results are provided in Table A.1. As shown, all the variables are statistically significant across all sub-periods. However, when dropping observations at the end of the study period (see Table A.2) the model shows bad fitness and the log currency to M1 becomes insignificant, while the rest of the causal variables are still significant.

Second, to check if the model suffers from an identification problem, model 4 was re-estimated without GDP per capita over the original study period from 1980 to 2017, and the results are provided in Table A.3. The findings showed that the model without GDP per capita impacts the badness of fit criteria, with CFI and TLI increased from 0.07, 0.051 to 0.102 and 0.65, respectively.

Third, all models were re-estimated using data in the standard form with a mean of zero and standard deviation of one to check for robustness of the coefficients estimated. From Table A.4, model 4 is still the optimal model. Furthermore, all the goodness and badness of fit criteria are better than the original results shown in Table A.5. However, as shown in the same table using data in growth form, GDP per capita is non-significant. Besides, the CFI and TLI criteria are high in comparison to the previous model estimations.

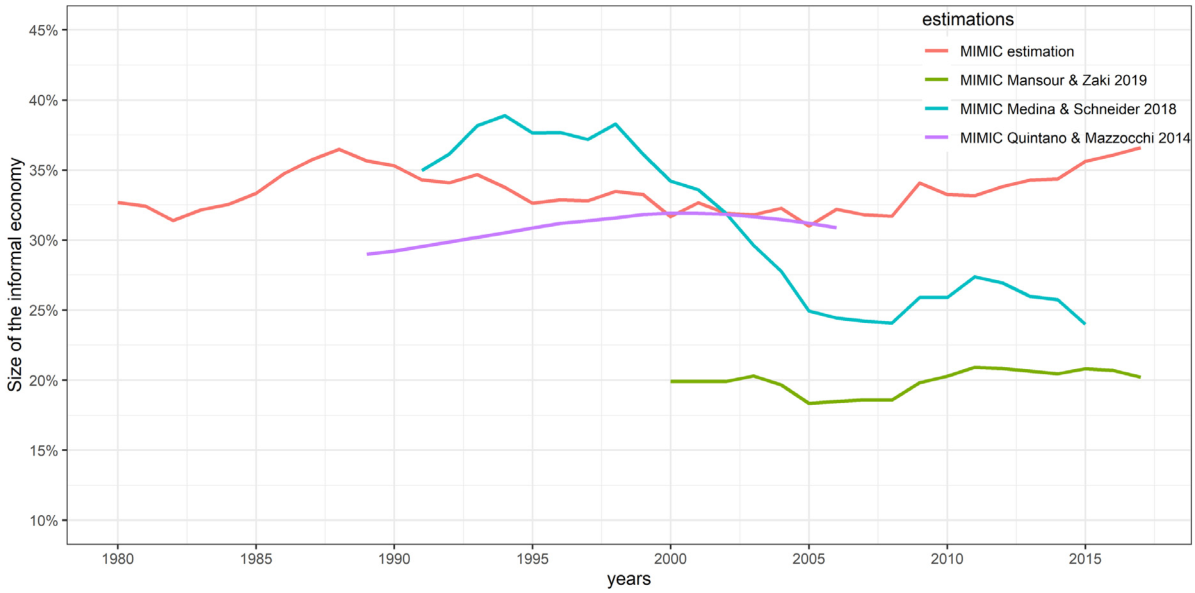

Fig. 3. The size and evolution of IE in Algeria in comparison to previous estimations.

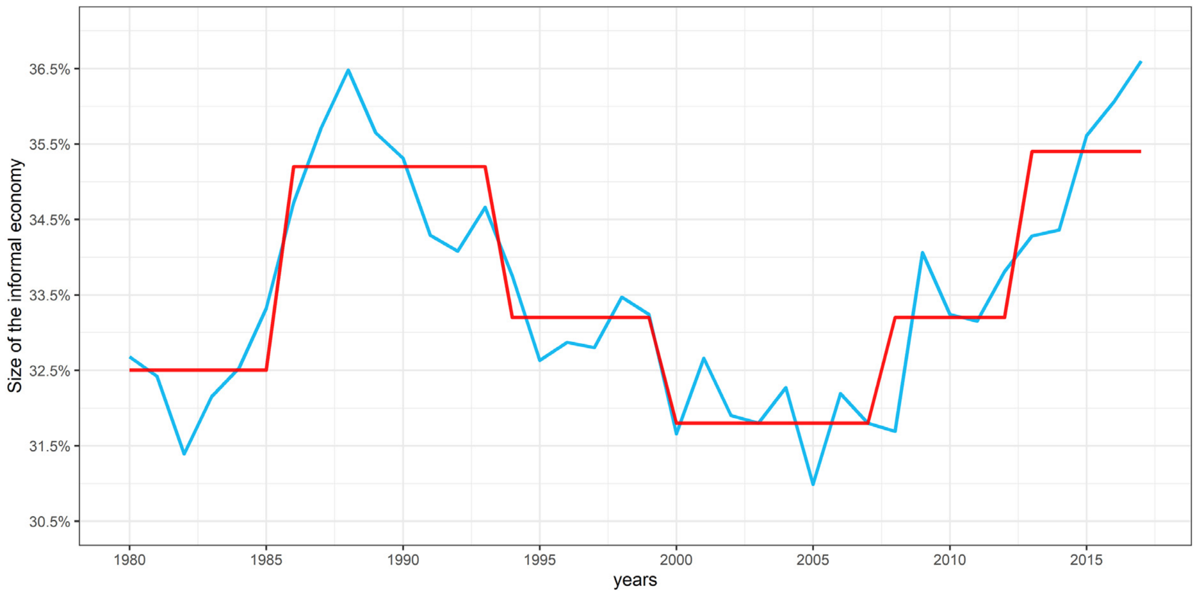

Fig. 4. IE estimations.

4. Interaction between informal and formal economy

4.1. Empirical estimation of the influence of the IE on the formal economy

To estimate the impact of the IE on the formal economy in Algeria, a growth model has been specified. This model consists of GDP per capita in constant USD ($gdppc_t$) as a dependent variable explained by independent variables used in general economic theory such as government final consumption (gfc_t), gross capital formation (gcf_t), human capital

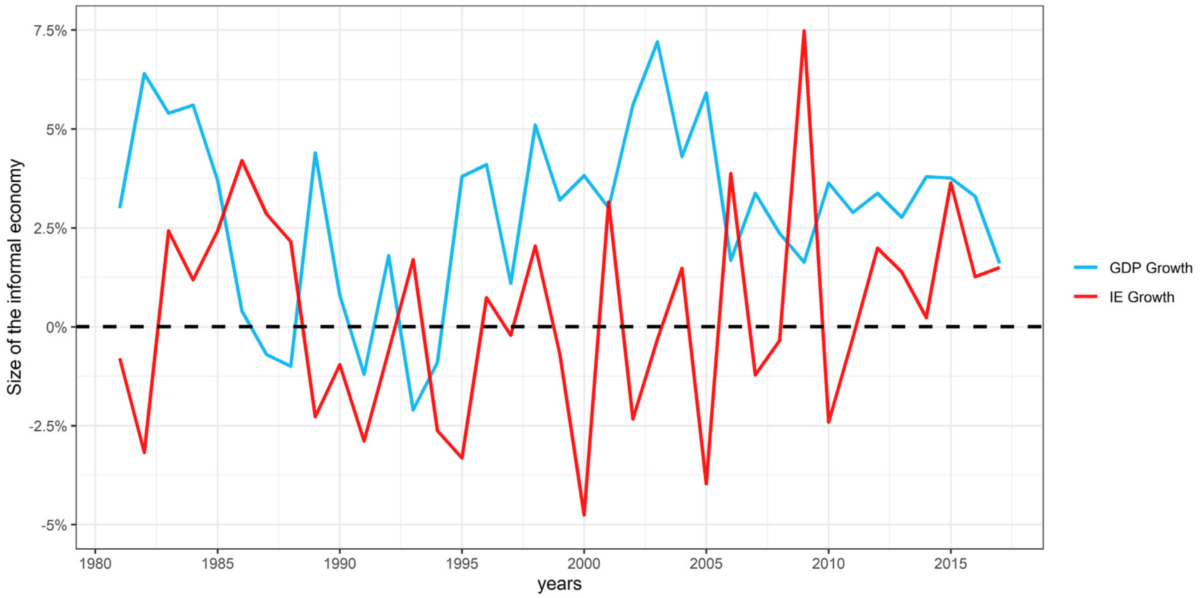

Fig. 5. Relationship between formal economy and IE in Algeria.

measured by secondary education (se_t), inflation (π_t)[1] and the size of the IE (ie_t).[2] In line with the economic theory, all the previous variables were log transformed. The model can be expressed mathematically as follows:

$$\ln gdppc_t = \beta_0 + \beta_1 \ln ie_t + \beta_2 \ln gfc_t + \beta_3 \ln gcf_t + \beta_4 \ln se_t + \varepsilon_t \tag{5}$$

where β_0 is the intercept, $\beta_i i = 1, 4$ are the regression coefficients, and ε_t is the error term.

Interestingly, in a recent study by Mughal and Schneider (2020), it was found that the IE has an asymmetric effect on the formal economy in both the long and short run.[3] Therefore, and to account for the asymmetric effect, the growth model is estimated using the autoregressive distributed lag (ARDL) method proposed by Pesaran et al. (1996) and subsequently was modified by Pesaran et al. (2001) by introducing the bounds testing approaches. Hence, Eq. (5) is written as follows:

$$\begin{aligned} D\ln gdppc_t = {} & \alpha_0 + \sum_{i=1}^{p1} \alpha_{1i} D\ln gdppc_{t-i} + \sum_{i=0}^{p2} \alpha_{2i} D\ln ie_{t-i} + \sum_{i=0}^{p3} \alpha_{3i} D\ln gfc_{t-i} + \sum_{i=0}^{p4} \alpha_{4i} D\ln gcf_{t-i} + \sum_{i=0}^{p5} \alpha_{5i} \ln se_{t-i} \\ & + \beta_1 \ln gdppc_{t-1} + \beta_2 \ln ie_{t-1} + \beta_3 \ln gfc_{t-1} + \beta_4 \ln gcf_{t-1} + \beta_{5i} \ln se_{t-1} + \varepsilon_t \end{aligned} \tag{6}$$

Where: D is the delay operator; α_0 is the constant; α_j means short-term dynamics; β_r are the long-term coefficients; ε_t Serially uncorrelated random errors with a mean of zero. The lag length for each variable of the ARDL (P1, P2, P3, P4, P5) is chosen by the Schwartz Information Criterion (SIC), assuming a maximum lag length of 4 lags.

The ARDL methodology requires variables to be integrated of order zero I(0) and one I(1) only. Hence, to ensure that none of the variables included in the growth model is integrated of an order higher than I(1), Augmented Dickey and Fuller (1979) (ADF) and Phillips and Perron (1988) (PP) unit root tests were used. To test the cointegration among the variables in Eq. (6), we employ the bounds cointegration approach of Pesaran et al. (2001), based on the partial F-test under the null of no cointegration H_0: $\beta_1 \neq \beta_2 \neq \beta_3 \neq \beta_4 \neq \beta_5 \neq 0$ against the alternative hypothesis of the existence of cointegration H_1: $\beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$. It is based on the F-statistics to accept or reject the null hypothesis. On the one hand, cointegration exists within the variables in the model when the F-statistic value is higher than the value of the upper bound, and no cointegration exists when the F-statistic is smaller than the lower bound. On the other hand, the results are inconclusive if the F-statistic falls between the upper and lower bounds.

According to the economic theory, a positive sign is expected for gross capital formation and secondary education. In contrast, a negative sign is expected for government consumption. The error correction term (ECT) is anticipated to be

[1] The inflation rate was excluded from the model due to the high autocorrelation in the residuals caused by it.

[2] Other variables such as employment, population, and inflation were dropped because their presence caused serial correlation, which led to low performance in the estimated models.

[3] We thank reviewer one for pointing out this issue.

Table 7
Unit root tests for the growth model variables over the period from 1980 to 2017.

Variables	At levels					
	ADF			PP		
	I	I&T	None	I	I&T	None
$\ln gdppc_t$	−1.0101	−1.5113	0.9529	−0.2455	−1.3032	1.1593
$\ln ie_t$	−1.2663	−1.3323	0.6978	−1.4250	−1.4344	0.7257
$\ln gfc_t$	0.3435	−2.0239	4.4844	0.2592	−1.6603	4.1285
$\ln gcf_t$	0.3955	−1.0543	1.4040	0.7487	−1.1065	1.9113
$\ln se_t$	−4.9755***	−2.2578	3.3918	−4.9755***	−2.3948	2.1489
$d\pi_t$	−2.6517*	−2.9412	−1.0369	−2.6450	−3.0013	−1.0468
	First difference					
	ADF			PP		
	I	I&T	none	I	I&T	none
$\ln gdppc_t$	−3.2339**	−3.2865*	−3.0911***	−3.2961**	−3.3391*	−3.1119***
$\ln ie_t$	−6.5556***	−2.8230	−6.5344***	−6.5295***	−6.5208***	−6.5116***
$\ln gfc_t$	−4.8994***	−4.9536***	−3.6895***	−4.8970***	−4.9086***	−3.6121***
$\ln gcf_t$	−3.8449**	−4.3983**	−3.5393***	−3.8512**	−4.3030**	−3.5189***
$\ln se_t$	−3.1475**	−3.8467**	−2.9143***	−3.1475**	−3.8467**	−2.9143***
$d\pi_t$	−8.5426***	−8.4394***	−8.6565***	−8.5426***	−8.4394***	−8.6565***

Notes: ***, **, * is the significance level at 1%, 5% and 10%, respectively. I represent the model with intercept only, and I&T represents the model with an intercept and trend. ADF is Augmented Dickey–Fuller Unit Root Test, the number of lags was determined using AIC criteria for maximum 9 lags to remove serial correlation in the residuals. PP is the Phillips–Perron unit root test, the test was conducted using Newey–West Bandwidth.

Table 8
Bounds co-integration test.

Calculated F-statistic		
6.6113	$K = 4$; $N = 34$	
	Critical value bounds	
significance	I(0) bound	I(1) bound
10%	2.46	3.46
5%	2.947	4.088
1%	4.093	5.532

statistically significant with an absolute value of less than one. As for our variable of interest, the literature is inconclusive about its effect on the formal economy; thus, a positive or negative effect is expected.

4.1.1. Econometric estimation

The results of ADF and PP unit root tests are displayed in Table 7. All the variables are found to be integrated of the first order I(1) at the 5% significance level, except for the secondary education variables, which is stationary at level I(0); hence, the ARDL bounds testing approach is suitable for our estimation. The bounds test results for cointegration are provided in Table 8. The calculated F-statistic of 6.6113 is greater than the upper bound critical value of 5.532 at the 1% significance level; thus, the long cointegration is established.

Panel C of Table 9 shows that the model has passed the serial correlation (Breusch–Godfrey Serial Correlation LM tests), normality (Jarque–Bera test), and heteroscedasticity (ARCH test). In addition, the CUSUM and CUSUM of squares graphs indicate that the model is stable and are provided in Appendix B. As shown in Table 9 panel A and B, gross capital formation and government final consumption have the right expected sign in the short and long run, whereas the second education variable has the wrong sign. In the long run, all the included variables have a highly significant impact on the formal economy at a 1% level, except for secondary education, which is statistically insignificant. In the short run, the results showed that government final consumption and gross capital formation are showing the expected signs.

For the essential variable in this study, the empirical results show that the IE has an asymmetric effect on the formal economy. In the short run, the IE has a significantly positive effect on the formal economy, while in the long run, it has a significantly negative effect on the formal economy. This finding can be explained by the lack of official employment opportunities. Keeping in mind that the majority of work opportunities in Algeria come from the state, individuals seek a primary income source in the IE, in the case of the unemployed, or a second income source for some individuals (primarily low-income state employees) to enhance their purchasing power. Hence, the income created in the IE is spent in the formal economy, establishing, in this way, a positive relationship in the short run. However, this effect reverses in the long run due to tax evasion that decreases the amount of tax revenue collected, which is considered one of the primary sources of government revenues.

4.2. Causality analysis

The Granger causality test is applied to investigate the causal relationship between the formal economy and the IE in Algeria (Giles, 1999; Soares and Afonso, 2019). This relationship is presented in Fig. 5 using the growth rate of the two

Table 9
Long and short run estimates.

Panel A: ARDL (4, 4, 4, 2, 0) long run coefficients				
Intercept	$\ln ie_t$	$\ln gfc_t$	$\ln gcf_t$	$\ln se_t$
6.5938***	−0.5695***	−0.1096***	0.2602***	−0.008
(13.3908)	(−4.0499)	(−4.3878)	(19.2246)	(−0.9652)

Panel B: ARDL (4, 4, 4, 2, 0) short run coefficients				
Lag order	0	1	2	3
$d\ln gdppc_t$		0.3991***	0.0989(0.8967)	0.2430**
		(3.5085)		(2.6496)
$d\ln ie_t$	−0.3124***	0.2731***	0.3092***	0.2128**
	(−4.6298)	(3.3350)	(4.1825)	(2.8129)
$d\ln gfc_t$	−0.1197***	0.0774*	−0.0651	−0.1385***
	(−3.6180)	(2.1732)	(−1.7173)	(−3.4630)
$d\ln gcf_t$	0.1348***	−0.0631		
	(7.4684)	(−2.2938)		
$d\ln se_t$				

Panel C: Diagnostic statistics tests						
ECM (−1)	Adj-R^2	DW Sta	F-Sta	χ^2_{LM}	χ^2_{JB}	χ^2_{ARCH}
−0.7986	0.9950	1.8073	371.0487	3.0141	1.83	0.32
[0.00]			[0.00]	[0.22]	[0.4]	[0.57]

Notes: ***, **, * is the significance level at 1%, 5% and 10%, respectively.
t-statistic in parentheses.
P-value in brackets.

Table 10
Granger causality test.

Variable	Direction of Granger causality	Variable	Results
$d\ln ie_t$	→	$d\ln gdppc_t$	0.3090 [0.5583]
$d\ln gdppc_t$	→	$d\ln ie_t$	4.2812 [0.0385]

Note: p-value in brackets.

Table 11
VAR estimations, over the period of 1980 to 2017.

Variable	$d\ln ie_t$	$d\ln gdppc_t$
$d\ln gdppc_{t-1}$	0.3854**	0.3455**
	(0.1862)	(0.1415)
	[2.0691]	[2.4424]
$d\ln ie_{t-1}$	0.0834	−0.0757
	(0.1794)	(0.1363)
	[0.4650]	[−0.5558]
c	−0.0031	0.0027
	(0.0061)	(0.0046)
	[−0.5057]	[0.5988]
Control variables		
$d\ln gfc_t$	−0.0019	−0.0534
	(0.1034)	(0.0785)
	[−0.0192]	[−0.68025]
$d\ln gcf_t$	−0.0300	0.1351
	(0.0455)	(0.0346)***
	[−0.0192]	(3.9050)
$d\ln se_t$	−0.1098**	−0.0317
	(0.0532)	(0.0404)
	(2.0622)	[−0.7847]
$d\pi_t$	0.015303**	−0.0044
	(0.0056)	(0.0044)
	[2.6877]	[−1.0350]
Adjusted R-squared	0.04	0.53
LM test	0.72	0.72

Notes: ***, **, * is the significance level at 1%, 5% and 10%, respectively.
standard deviation in parentheses.
t-statistic in brackets.

economies. The main idea of the causality test is that if a one-time series (say x) is Granger causative, another time series (say y) should precede it. The two time series investigated must be stationary when using the Granger causality, which is an essential condition (Granger, 1969). In our case, we examine causality between the IE measured by the obtained

MIMIC estimation in Section 3 as a percentage of official GDP and the formal economy measured with GDP per capita in constant USD. The causality model included additional control variables mentioned in the subsection above. All the included variables are in logarithmic form. As a first step, stationarity properties are checked from Table 7; the results suggest that all variables are stationary after taking the first difference. For the second step, AIC and FPE information criteria are used to select the optimum number of lags; based on these two tests, the optimum lag number is 1.

Table 10 presents the results of the Granger causality. The arrow shows the direction of causality tested. It is found that there is unidirectional causality in which GDP per capita Granger causes the IE. This result was found in Giles (1999) in the case of New Zealand. As Table 11 shows, GDP per capita with one lag[4] positively impacts the size of IE, and this impact is significant at the 5% level. This implies that the IE economy follows the formal one through the cycle. The estimated positive sign suggests that the two economies are complementary. This finding is significant for policymakers because the implementation of expansionary economic policies will stimulate the size of the IE, e.g., according to Giles (1999), when implying an expansionary fiscal policy by reducing tax rates, ceteris paribus, the size of the IE will decrease. Hence, policymakers must consider that individuals will be motivated to switch to the IE when applying contractionary fiscal policy. More importantly, they must consider that the poor people working in the IE will be affected, and their living conditions will deteriorate.

5. Conclusions and policy implications

The quantitative analyses revealed that the average IE in Algeria between 1980 and 2017 was 33.48% of official GDP, and it has continued to increase over the past fifteen years. Based on the MIMIC approach, it is found that the key cause of the IE is the tax burden followed by the agricultural sector and the quality of institutions.

The agricultural sector is the second most important determinant. To our knowledge, it must be noted that this variable has not been accounted for in other studies. This finding is particularly important because, in the Algerian case, the agricultural sector is considered vital to the economic growth vision made by the government, and for this, it is heavily funded by the state. To stimulate and develop this sector, the Algerian government has adopted various economic policies. According to the Minister of Agriculture, more than 20, 000 investment projects costing 200 billion DA were carried out from 2010 to 2016 in the agricultural sector and later 60,000 projects were financed by the National Agency for the Support of Youth Employment (NASYE), with a 7% reduction in taxes on materials used in the agricultural sector, besides debt clearance in 2008 and 2014 by presidential order. However, the results showed that the IE is reducing the impact of these economic policies, and almost half of their impact is lost to the IE.

The quality of bureaucratic institutions is found to positively impact the progress of the IE. This reverse impact is explained by the high levels of corruption in Algeria. It was argued that the insignificant relation between unemployment and the IE in the Algerian case is due to the heterogeneous unemployed labour force. This result means that engaging in informal activities depends on the situation of the IE. Another relative finding is that individuals try to recompense for lost revenue in declared activities in times of recession by engaging in the IE. This is reflected by the negative relation between GDP per capita and the IE.

Concerning the relationship between the formal economy and the IE, it was found that the formal economy Granger causes the IE, and this cause-and-effect relationship is unidirectional. It means that when policymakers propose policies to stimulate the formal economy, a portion of stimulus funds are lost to the IE, thus reducing policy effectiveness. Unfortunately, due to the lack of data and its periodicity, some significant theoretical causes were omitted from the study, whereas the empirical part suffers from a small sample size.

In terms of policy implications, in order to reduce the size of the IE, the Algerian government should adopt a mixed economic policy to target the various sources of the IE by reducing the tax burden; decreasing the informal labour in the agricultural sector; lowering the degree of bureaucratic procedures and fighting corruption in the government administration system and enhancing the economic welfare of its citizens.

For instance, taxes should depend on the economic situation; for example, the tax burden should be reduced in times of recession. In this way, policymakers can reduce people's motivation to engage in informal activities. Increasing people's confidence in tax authorities by ensuring more transparency in managing tax resources, sensitizing citizens to tax goals and most importantly, combating corruption at the level of tax administration would also help. As the empirical results suggest that corruption is an implicit cause of the IE, combating this phenomenon is a key factor in reducing the size of the IE in Algeria.

Another important factor in reducing the size of the IE is the use of electronic money. Unfortunately, until today, all economic activities have been conducted using cash. If the Algerian government succeeds in generalizing electronic transactions and at the same time withdraws cash from the economy, it will reduce the number of transactions made in the IE, thus reducing its size. In addition, the Algerian government must make in motion Law No. 08/96 of December 1996 regulating the conditions and how to establish money exchange offices to attract the largest possible number of informal money changers by establishing private and official money exchange offices with payment of taxes and social security contributions. By doing this, the government provides new job opportunities and ensures a vital resource of taxes and foreign money supply.

4 The same result is found when using other measures of the formal economy and changing control variables.

The authorities must review the low official salaries and wages, and this is a fundamental issue because it has been years since the last time the Algerian government made a significant change in the official salaries. For example, according to a survey on teachers' salaries conducted by the higher education research centre of the University of Chicago, the average Algerian teacher's salary is $US345, which is considered very low compared to other MENA countries like Kuwait $2890. However, in a study conducted by CASI (Confédération Algérienne des Syndicats Indépendants), the minimum average monthly wage needed to live in Algeria was found to be $600, and this is considered more than intuitive to search for another income source. Therefore, keeping in mind that the study found a negative effect of personal income on the size of the IE, increasing official salaries and wages will reduce individuals' intention to work in the IE, hence its size.

Finally, with the established positive relationship between the agricultural sector and the IE size, and with the Algerian endeavour to stimulate this specific sector, the Algerian government faces a difficult challenge because only a portion of these adopted expansionary economic policies will find their way in increasing the informal activities, hence reducing their efficiency. Therefore, it is necessary to monitor these economic policies and consider the effect that the IE has on them.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Robustness checks

See Tables A.1–A.5.

Table A.1
Sample variation by dropping observations at the beginning of the period.

	Period from 1981 to 2017 (37 obs.) Model 4	Period from 1985 to 2017 (33 obs.) Model 4	Period from 1990 to 2017 (28 obs.) Model 4	Period from 1995 to 2017 (23 obs.) Model 4
Determinants				
Tax burden as % of GDP	0.499***	0.533***	0.485***	0.432***
	(4.354)	(5.069)	(4.097)	(2.995)
Unemployment rate				
Added agriculture value	0.494***	0.451***	0.444***	0.486***
as % of GDP	(5.544)	(5.358)	(4.114)	(4.122)
Quality of institutions	0.344***	0.343***	0.342***	0.128*
	(4.562)	(5.705)	(4.821)	(1.824)
GDP per capita growth	−0.269**	−0.272***	−0.198*	−0.255
rate	(−2.648)	(−2.858)	(−1.828)	(−1.548)
Indicators				
Log current GDP	−1	−1	−1	−1
Log currency to M1	0.34**	0.377**	0.369**	0.501***
	(2.296)	(2.265)	(2.159)	(3.626)
Employment to the working population				
Model fit indicators				
Chi-square	4.73	4.26	5.53	6.863
	[0.32]	[0.81]	[0.766]	[0.816]
RMSEA	0.07	0.04	0.117	17.6
	[0.27]	[0.24]	[0.275]	[0.177]
SRMR	0.051	0.058	0.77	0.074
CFI	0.98	0.995	0.969	0.916
TLI	0.95	0.989	0.93	0.811
Degrees of freedom	14	14	14	14
R-square	0.76	0.80	0.811	0.73

Note: ***, **, * is the significance level at 1%, 5% and 10%, respectively.
Z-values in parentheses.
P-value in brackets.

Table A.2
Sample variation by dropping observation at the end of the period.

	Period from 1981 to 2017 (37 obs.) Model 4	Period from 1981 to 2013 (33 obs.) Model 4	Period from 1981 to 2008 (28 obs.) Model 4	Period from 1995 to 2017 (23 obs.) Model 4
Determinants				
Tax burden as % of GDP	0.499***	0.469***	0.466***	0.498***
	(4.354)	(4.051)	(3.297)	(3.482)
Unemployment rate				
Added agriculture value	0.494***	0.45***	0.41***	0.433***
as % of GDP	(5.544)	(4.691)	(3.518)	(3.192)
Quality of institutions	0.344***	0.363***	0.438***	0.128***
	(4.562)	(4.654)	(4.475)	(4.715)
GDP per capita growth	−0.269**	−0.272**	−0.198*	−0.323
rate	(−2.648)	(−2.355)	(−2.002)	(−2.174)
Indicators				
Log current GDP	−1	−1	−1	−1
Log currency to M1	0.34**	0.2	−0.051	−0.008
	(2.296)	(1.245)	(−0.314)	(−0.046)
Employment to the working population				
Model fit indicators				
Chi-square	4.73	6.073	4.517	5.224
	[0.32]	[0.97]	[0.34]	[0.27]
RMSEA	0.07	0.124	0.06	0.13
	[0.27]	[0.31]	[0.292]	[0.382]
SRMR	0.051	0.06	0.06	0.074
CFI	0.98	0.941	0.977	0.93
TLI	0.95	0.868	0.949	0.841
Degrees of freedom	14	14	14	14
R-square	0.76	0.67	0.603	0.65

Note: ***, **, * is the significance level at 1%, 5% and 10%, respectively.
Z-values in parentheses.
P-value in brackets.

Table A.3
Model 4 without GDP per capita growth estimations over the period from 1981 to 2017.

	Model 4 Original estimations	Model 4 without GDP per capita
Determinants		
Tax burden as % of GDP	0.499***	0.543***
	(4.354)	(4.286)
Unemployment rate		0.035
		(0.266)
Added agriculture value as % of GDP	0.494***	0.444***
	(5.544)	(3.948)
Quality of institutions	0.344***	0.334***
	(4.562)	(3.302)
GDP per capita growth rate	−0.269**	
	(−2.648)	
Indicators		
Log current GDP	−1	−1
Log currency to M1	0.34**	0.35**
	(2.296)	(2.108)
Employment to the working population		
Model fit indicators		
Chi-square	4.73	5.95
	[0.32]	[0.8]
RMSEA	0.07	0.102
	[0.27]	[0.26]
SRMR	0.051	0.065
CFI	0.98	0.956
TLI	0.95	0.901
Degrees of freedom	14	14
R-square	0.76	0.743

Note: ***, **, * is the significance level at 1%, 5% and 10%, respectively.
Z-values in parentheses.
P-value in brackets.

Table A.4
Re-estimation using standard data with mean of zero and standard deviation of one.

	Model 1 (5-1-3)	Model 2 (4-1-3)	Model 3 (5-1-2)	Model 4 (4-1-2)	Model 5 (3-1-2)
Determinants					
Tax burden as % of GDP	0.469***	0.465***	0.510***	0.509***	0.57***
	(4.448)	(4.344)	(4.478)	(4.355)	(4.282)
Unemployment rate	0.067		0.088		0.047
	(0.631)		(0.771)		(0.349)
Added agriculture value as % of GDP	0.426***	0.442***	0.485***	0.514***	0.483***
	(4.547)	(5.264)	(4.831)	(5.650)	(4.065)
Quality of institutions	0.335***	0.323***	0.364***	0.35***	0.349***
	(4.395)	(4.45)	(4.599)	(4.589)	(3.378)
GDP per capita growth rate	−0.257**	−0.246**	−0.288**	−0.227***	
	(−2.528)	(−2.473)	(−2.788)	(−2714)	
Indicators					
Log current GDP	−1	−1	−1	−1	−1
Log currency to M1	0.263*	0.266**	0.358**	0.36**	0.378**
	(1.875)	(1.927)	(2.327)	(2.387)	(2.256)
Employment to the working population	0.093	0.097			
	(0.83)	(0.848)			
Model fit indicators					
Chi-square	21.21	22.47	5.98	4.16	4.78
	[0.01]	[0.91]	[0.308]	[0.38]	[0.31]
RMSEA	0.182	0.192	0.075	0.033	0.066
	[0.02]	[0.017]	[0.25]	[0.26]	[0.353]
SRMR	0.09	0.087	0.058	0.049	0.061
CFI	0.74	0.756	0.975	1	0.98
TLI	0.575	0.594	0.946	0.99	0.959
Degrees of freedom	26	19	20	14	9
R-square	0.636	0.64	0.781	0.80	0.84

Note: ***, **, * is the significance level at 1%, 5% and 10%, respectively. Z-value in parentheses. P-value in brackets. Degrees of freedom $= 0.5(p+q)(p+q+1) - j$, where p is the number of causal variables, q is the number of indicator variables, j is the number of free parameters.

Table A.5
Model 4 re-estimations with growth data from 1981 to 2017.

	Growth data		Growth data	The selected model estimations
	Model 4	Model 4 with employment	Model 4 without GDP per capita growth	Model 4
Determinants				
Tax burden as % of GDP	0.582***	0.539***	0.585***	0.499***
	(4.234)	(4.261)	(3.815)	(4.354)
Unemployment rate	0.129	0.114	0.08	
	(0.975)	(0.903)	(0.713)	
Added agriculture value as % of GDP	0.425***	0.374***	0.441***	0.494***
	(3.889)	(3.757)	(4.184)	(5.544)
Quality of institutions	0.308***	0.29***	0.283**	0.344***
	(2.898)	(2.981)	(2.463)	(4.562)
GDP per capita growth rate	0.138	0.146		−0.269**
	(1.144)	1.262)		(−2.648)
Indicators				
Log current GDP	−1	−1	−1	−1
Log currency to M1	0.40**	0.337**	0.42**	0.34**
	(2.814)	(2.462)	(2.417)	(2.296)
Employment to the working population		0.144		
		(0.993)		
Model fit indicators				
Chi-square	8.22	20.472	5.77	4.73
	[0.14]	[0.92]	[0.21]	[0.32]
RMSEA	0.11	0.153	0.11	0.07
	[0.26]	[0.07]	[0.26]	[0.27]
SRMR	0.07	0.09	0.06	0.051
CFI	0.93	0.79	0.96	0.98
TLI	0.85	0.65	0.91	0.95
Degrees of freedom	21	19	14	14
R-square	0.81	0.676	0.80	0.76

Note: ***, **, * is the significance level at 1%, 5% and 10%, respectively. Z-value in parentheses. P-value in brackets. Degrees of freedom $= 0.5(p+q)(p+q+1) - j$, where p is the number of causal variables, q is the number of indicator variables, j is the number of free parameters.

Appendix B. CUSUM and CUSUM of squares

See Figs. B.1 and B.2.

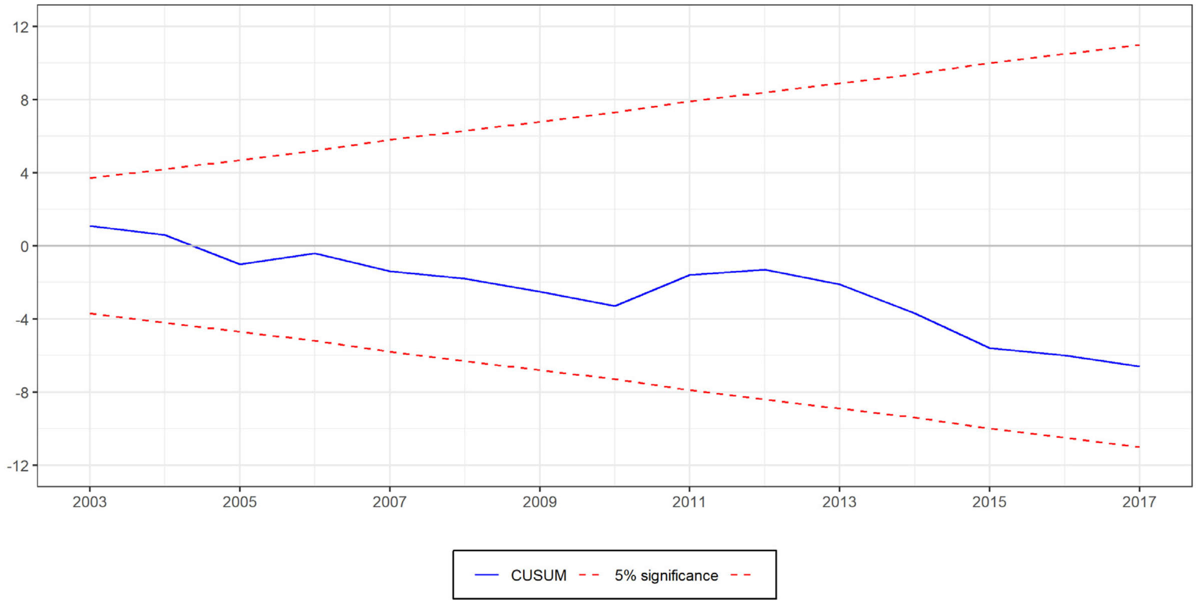

Fig. B.1. CUSUM.

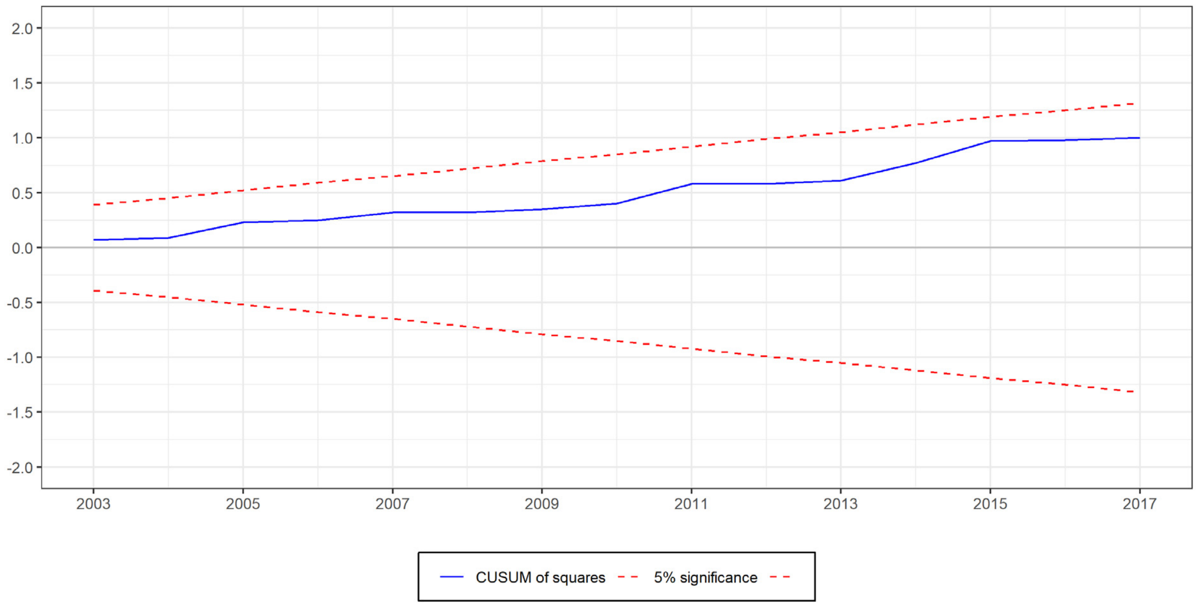

Fig. B.2. CUSUM of squares.

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